

Arunabha Majumder

Mechanical Design Engineer | 3D CAD | Prototyping | Electromechanical Systems

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September 27, 2026

Hiring Team, R&D Mechanical Design
Eaton

Application: Mechanical Design Engineer – UPS Systems

Dear Hiring Team,

I am applying for the Mechanical Design Engineer position working on UPS systems and accessories. I hold a B.Tech in Mechanical Engineering and an M.Tech in Mechatronics, and I am finishing my PhD in Mechatronics and Robotics at Aalborg University. For the past three years I have designed, documented, built and tested electromechanical products in R&D, and I want to bring that experience to power management products at Eaton.

In my PhD, I designed a hybrid-actuated wearable device from concept to tested hardware. I created the SolidWorks 3D models and drawings, iterated the design on FEA results for stiffness, strength and weight, and built the prototypes hands-on from machined and additively manufactured parts. I integrated the motor, encoder, force sensors and DAQ, and built custom test rigs to validate the design. The hybrid actuation cut energy consumption by more than 25% and won the Best Paper Award at IFToMM ISRM 2026.

I enjoy analysing and solving product issues, and I keep cost in mind while doing it. On a 5-bar parallel robot, I removed an external force/torque sensor by estimating force from the actuator itself, which reduced system cost, mass and integration complexity. I also validated my analytical stiffness models against physical tests to within 10% RMS error before committing to fabrication.

I work well in R&D project teams. My projects combine mechanical design, electronics, controls and human testing, and I have worked in research environments in both India and Denmark with English as my working language. I am a Certified SolidWorks Associate and have published my work in peer-reviewed venues.

My hands-on manufacturing experience so far is mainly with machined and additively manufactured parts. Sheet metal, die-cast and injection-moulded design, together with DFSS tools, is where I want to grow next, and Eaton's UPS R&D team is the right place to do it.

I would welcome the opportunity to discuss how I can contribute to your team. References are available upon request.

Kind regards,
Arunabha Majumder